IPL DATA ANALYSIS PROJECT

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Institution	:	Mohamed Sathak Engineering College (Anna University)
Programme	:	B.Tech Information Technology (2024–2028)
Project Type	:	End-to-End Data Analysis — Self Project & Portfolio
Dataset	:	IPL Ball-by-Ball Data	2008–2026	Cricsheet.org
Total Records	:	**1,243 Matches	295,732+ Deliveries	1,000+ JSON Files**
Tools Used	:	Python · MySQL · Power BI · DAX · Jupyter Notebook
Project Status	:	Completed

ABSTRACT

This project presents a comprehensive end-to-end data analysis of the Indian Premier League (IPL) spanning 18 seasons from 2008 to 2026. Working with a massive dataset of over **295,732 ball-by-ball delivery records** across **1,243 matches**, sourced from over **1,000 individual JSON files** from Cricsheet.org, this analysis uncovers key patterns in player performance, team dominance, toss impact, and venue behaviour. The project follows a complete data pipeline — from raw JSON extraction using Python, through MySQL database modelling using a Star Schema, to an interactive Power BI dashboard with DAX measures — making it a strong demonstration of real-world data analyst skills.

PROBLEM STATEMENT

Despite the IPL generating enormous amounts of match data every season, this data remains largely unexplored in a structured analytical form. This project aims to perform a comprehensive analysis on IPL data to answer 10 key business questions around player performance, team win patterns, toss impact, venue behaviour, and season-wise scoring trends — delivering insights through an interactive two-page Power BI dashboard.

10 Analysis Questions:

No.	Category	Question
Q1	BATTING	Who are the top 10 run-scorers with their batting average and strike rate?
Q2	BATTING	Which batsmen hit the most sixes and fours across all IPL seasons?
Q3	BOWLING	Who are the top 10 wicket-takers with economy rate and bowling average?
Q4	BOWLING	Which bowlers are the most economical in death overs (overs 16-20)?

No.	Category	Question
Q5	TEAM	Which team has the highest win count across all IPL seasons?
Q6	TEAM	Which team performs better while chasing vs defending?
Q7	MATCH	Does winning the toss increase the chances of winning the match?
Q8	VENUE	What is the average first-innings score per venue?
Q9	TRENDS	How has the average team score evolved from 2008 to 2026?
Q10	PLAYER	Which players win the most Player of the Match awards?

DATA SOURCE & MASSIVE SCALE

The dataset was sourced from **Cricsheet.org**, the most authoritative repository of ball-by-ball cricket data. The IPL dataset came as a ZIP archive containing **over 1,000 individual JSON files** — one file per match —

each containing deeply nested structures with match metadata, innings data, over-by-over delivery information, wicket details, extras, and fielding information. This required building a custom Python parser to loop through all JSON files, flatten the nested structure, and combine everything into two structured CSVs

TotalJSON Files	Total Matches	TotalDeliveries	IPLSeasons
1,000+	1,243	295,732+	18 (2008-2026)

KEY CHALLENGES & SOLUTIONS

1. Team Name Inconsistencies

One of the most significant data quality issues encountered was that several IPL franchises have changed their names over the years. The dataset contained **multiple versions of the same team** appearing as separate entries, causing duplicate bars in every chart and incorrect win counts. The affected franchises were: Delhi Daredevils / Delhi Capitals, Kings XI Punjab / Punjab Kings, Royal Challengers Bangalore / Royal Challengers Bengaluru, and Gujarat Lions / Gujarat Titans. These were resolved by running targeted **SQL UPDATE statements** across all tables (matches, dim_match, dim_team, fact_deliveries) to standardise all franchise names to their current form.

2. Venue Name Inconsistencies

Stadium names presented an even larger challenge — the same ground appeared under **30+ different name variations** across different seasons. For example, the Wankhede Stadium appeared as 'Wankhede Stadium' and 'Wankhede Stadium, Mumbai'; the MA Chidambaram Stadium appeared as 'MA Chidambaram Stadium', 'MA Chidambaram Stadium, Chepauk', and 'MA Chidambaram Stadium, Chepauk, Chennai'; and Feroz Shah Kotla was also listed as 'Arun Jaitley Stadium' and 'Arun Jaitley Stadium, Delhi' following its renaming. Over **15 venue groups** required standardisation using SQL UPDATE queries targeting all three relevant tables to ensure accurate venue-based analysis.

3. Processing 1,000+ JSON Files

The raw data from Cricsheet came as individual JSON files — one per match — with deeply nested structures. Each file required parsing match metadata, iterating through innings, overs, and individual deliveries, extracting wicket details, extras breakdown, and fielder information. A custom Python script was written to loop through all JSON files, parse each one, and combine the results into two flat CSVs (matches.csv and deliveries.csv) ready for database import.

4. Season Format Issues

The 2008 IPL season (which started in 2007) was stored as '2007/08' in some records and as 2007 as an integer in others, causing the season slicer in Power BI to miss 2008 entirely. This was resolved with a targeted SQL UPDATE to standardise the season year to 2008 across all affected tables.

5. Large Data Import into MySQL

The deliveries.csv file at 29.9 MB was too large for the MySQL Workbench Table Data Import Wizard, which threw an 'Unhandled exception' error. This was solved by using SQLAlchemy's to_sql() method in Python to push the data directly from a Pandas DataFrame into MySQL — a more robust and scalable approach than the GUI wizard.

METHODOLOGY & PIPELINE

Step	Tool	Task	Output
1	**Python (Jupyter)**	Parse 1,000+ JSON files Extract & flatten nested data	matches_raw.csv deliveries_raw.csv
2	**Python (Pandas)**	Clean nulls, fix types, standardise text Handle outliers and duplicates	matches_clean.csv deliveries_clean.csv
3	**MySQL (Workbench)**	Load CSVs into MySQL Create Star Schema tables Fix team/venue names with SQL	ipl_db database 5 tables (Star Schema)
4	**SQL**	Write JOIN queries Answer 10 analysis questions	Query results ipl_analysis.sql
5	**Python (Matplotlib)**	Visualise SQL results Generate 8 analysis charts	PNG chart files Jupyter Notebook
6	**Power BI + DAX**	Connect to MySQL Build relationships Write DAX measures Create 2-page dashboard	ipl_dashboard.pbix Interactive dashboard

DATA MODELLING — STAR SCHEMA

The cleaned data was loaded into a MySQL database (ipl_db) and modelled using a **Star Schema** — the industry standard for analytical databases. The schema consists of one central fact table connected to four dimension tables via foreign key relationships.

Table	Type	Rows	Key Columns
fact_deliveries	Fact	295,732+	match_id, inning, over, ball, batter, bowler, runs_batter, is_wicket
dim_match	Dimension	1,243	match_id, season, date, venue, team1, team2, toss_winner, winner
dim_player	Dimension	700+	player_name
dim_team	Dimension	15	team_name
dim_venue	Dimension	64	venue, city

KEY FINDINGS & INSIGHTS

Question	Finding
Q1 — Top Run Scorer	Virat Kohli leads with 9,000+ IPL runs — the only batter to cross this milestone
Q2 — Most Sixes	CH Gayle holds the record for most sixes in IPL history — true to his 'Universe Boss' reputation
Q3 — Top Wicket Taker	Bhuvneshwar Kumar leads the wicket-takers chart with consistent performance over seasons
Q4 — Death Over Economy	Jasprit Bumrah and Rashid Khan are the most economical bowlers in the critical death overs
Q5 — Most Wins	Mumbai Indians dominate with the highest win count, followed by Chennai Super Kings
Q6 — Chase vs Defend	Teams winning the toss and choosing to field win 53% of the time
Q7 — Toss Impact	Toss winners win 51.56% of matches — slightly above 50%, proving toss has a marginal mpact
Q8 — Highest Scoring Venue	Narendra Modi Stadium, Ahmedabad and Wankhede Stadium, Mumbai
Q9 — Season Trend	Average first innings scores have risen from ~150 in 2008 to nearly 190+ in recent seasons
Q10 — POTM Awards	AB de Villiers leads Player of the Match awards, followed closely by CH Gayle and Virat Kohli

DAX MEASURES

Measure	Formula	Result
Batting Average	DIVIDE(SUM(runs_batter), SUM(is_wicket))	**25.95**
Economy Rate	DIVIDE(SUM(runs_total) * 6, COUNT(ball))	**8.15**
Toss Win %	DIVIDE(COUNTROWS(toss_winner = winner), COUNTROWS(valid_matches))	**51.56%**
Avg Score Per Match	DIVIDE(SUM(runs_total inning=1), DISTINCTCOUNT(match_id inning=1))	**168.57**

POWER BI DASHBOARD

The final deliverable is a **two-page interactive Power BI dashboard** built using the Teal & Light Green colour theme, connected directly to the MySQL ipl_db database. The dashboard includes dynamic season slicers on both pages, allowing users to filter all visuals by year.

Page	Title	Visuals Included
Page 1	**Team & Match Overview**	4KPI Cards(Matches, Seasons, Teams, Toss Win %) · Team WinCountBar Chart · Toss Winners C
Page 2	**Player & Venue Analysis**	2KPI Cards(AvgScore, Economy Rate) · Top 10 Run Scorers · Top10 WicketTakers · Player of the Match

SKILLS DEMONSTRATED

Skill	Application in Project
Python (Pandas, JSON)	Parsed 1,000+ JSON files, cleaned 295,732 records, handled nulls, type mismatches, outliers
SQL (MySQL)	Star Schema design, CREATE TABLE, INSERT, UPDATE, JOIN queries, reserved word handling

Skill	Application in Project
Power BI	MySQL connection, relationship modelling, 2-page interactive dashboard, slicers
DAX	Batting Average, Economy Rate, Toss Win %, Avg Score Per Match custom measures
Data Modelling	Star Schema with 1 fact table + 4 dimension tables, foreign key relationships
Data Cleaning	Team name standardisation, venue deduplication, season format fixing, SQL UPDATE
Data Visualisation	Matplotlib charts, Power BI bar/donut/line charts, KPI cards
Problem Solving	Fixed large file import issues, JS-rendered website blocking, MySQL reserved words